

**UNIVERSITE CATHOLIQUE D'AFRIQUE
CENTRALE**

FACULTES DES SCIENCES SOCIALES ET
DE MANAGEMENT

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**Descriptive Statistics
Continuous Assessment**

Second Semester - Academic Year 2025-2026

(Time: 2 hours)

Courses Questions: (05 points)

Define the following terms: random variable, descriptive statistics, Numerical variables, categorical variables, and measures of central tendency.

Exercise I:

Part A: (05 points)

Consider the following scores obtained by 50 students in the descriptive statistics' continuous assessment:

8, 7, 5, 6, 2, 8, 7, 6, 7, 3, 9, 10, 4, 5, 5, 4, 6, 7, 4, 8, 2, 2, 11, 10, 9, 3, 5, 6, 11, 7, 9, 8, 2, 4, 7, 9, 8, 4, 6, 7, 8, 3, 6, 7, 9, 10, 9, 11, 6, 8

- 1- What type of variable represents the distribution of scores?
- 2- Create a statistical table containing all the data
- 3- Determine the range, the arithmetic mean, the mode, and the variance
- 4- Create a bar chart representing the distribution of scores

Exercise II: (05 points)

In the following contingency table, X represents a country's trade balance and Y the exchange rate.

X \ Y	Y		
	[0,10[	[10,20[	[20,30[
[0,4[	0	40	0
[4,8[	20	0	20

- 1- Determine the marginal distributions of X and Y
- 2- Calculate the means and variances of these marginals' distributions

Exercise III: (05 points)

The following table gives the values of Consumption (Y) by a household in the city of Yaoundé according to annual income (X in thousands of FCFA)

Income (X)	42	65	66	82	86	130
Consumption(Y)	38	64	63	74	79	107

- 1- Calculate average income and average consumption
- 2- Determine the covariance between X and Y noted $COV(X, Y)$ as well as the Variance of X, $V(X)$.
- 3- Knowing that the regression line equation of consumption (Y) as a function of income (X) is of the form $Y = ax + b$, where $a = \frac{COV(X,Y)}{V(X)}$ and $b = \bar{Y} - a\bar{X}$, derive a and b.

Dr. NGAMBOE